

Hemostasis, Thrombosis & Bleeding Disorders - Factoid Sheet

Week 7 Hematology • Lecture 6 • Savage • memorize-first companion

ScholarRx: [Coagulation Cascade](#) • [Bleeding Disorders](#) • [Tests for Hemostasis](#) • [Hemophilia](#) • [TMA](#) • [Thrombotic Disorders](#)

Primary = platelet plug / mucosal. Secondary = fibrin / deep delayed bleeding.

Primary hemostasis

- **Adhesion:** vWF-GpIb
- **Activation:** ADP-P2Y₁₂, TXA₂, granules
- **Aggregation:** GpIIb/IIIa-fibrinogen
- vWF carries factor VIII

Platelet disorders

- **Glanzmann:** GpIIb/IIIa; ristocetin normal
- **Bernard-Soulier:** GpIb; giant platelets; ristocetin abnormal
- **ITP:** isolated low platelets, high MPV, normal PT/aPTT
- **vWD:** adhesion defect +/- low VIII

Coagulation cascade + tests

Item	Memorize
Intrinsic	XII -> XI -> IX + VIII; aPTT
Extrinsic	TF/III + VII -> X; PT/INR
Common	X + V -> II (thrombin) -> I (fibrin); XIII crosslinks
Vitamin K	II, VII, IX, X + C, S
Mixing study	Corrects = deficiency; fails = inhibitor
Factor XIII	Bleeding can occur with normal PT/aPTT
Factor XII	Can prolong aPTT without bleeding

Hemophilia B: factor IX, X-linked, deep/joint bleeding, PT normal, aPTT prolonged, mixing corrects if no inhibitor, IX assay low.

Hemophilia + vWD

	Hemophilia A/B	vWD
Bleeding	Deep, joints, muscles, delayed	Mucocutaneous; heavy menses/epistaxis
Platelets	Normal	Usually normal
PT	Normal	Normal
aPTT	High	Normal or high if VIII low
Key test	Factor VIII or IX assay	vWF antigen + activity/ristocetin

MAHA + Thrombophilia Factoids

DIC vs TTP vs HUS

	DIC	TTP	HUS
Schistocytes	Yes	Yes	Yes
Platelets	Low	Low	Low
PT/aPTT	High	Normal	Usually normal
Key clue	Sepsis/APL/trauma/OB; D-dimer up, fibrinogen down	ADAMTS13; neuro; PLASMIC	Child + bloody diarrhea + renal failure
Tx	Treat cause/support	PLEX now	Supportive (typical)

Natural anticoagulants

- **Antithrombin:** IIa/Xa especially; heparin accelerates it
- **Protein C + S:** inactivate Va/VIIIa
- C/S are vitamin K-dependent

Inherited thrombophilia

- **FVL:** factor V resists protein C
- **Prothrombin G20210A:** too much prothrombin
- **C/S deficiency:** failure to turn off V/VIII
- Don't test C/S during acute clot/warfarin if avoidable

HIT

Heparin + PF4 IgG -> platelet activation -> thrombosis + falling platelets. Low platelets, but the disease is a **clotting** emergency.

- **4 Ts:** Thrombocytopenia, Timing, Thrombosis, oTher causes.
- High suspicion: **stop all heparin** and start a non-heparin anticoagulant.
- PF4 ELISA = sensitive screen; SRA = functional confirmation.
- Don't rush into warfarin during acute HIT.

If you see X -> think Y

- Mucosal + normal platelets + abnormal ristocetin -> **vWD**
- Giant platelets + no ristocetin response -> **Bernard-Soulier**
- Normal ristocetin + aggregation failure elsewhere -> **Glanzmann**
- Hemarthrosis + isolated high aPTT correcting on mix -> **Hemophilia A/B**

- Schistocytes + low Plt + normal coag -> **TTP/HUS**
- Schistocytes + low Plt + high PT/aPTT -> **DIC**
- Thrombosis + heparin day 5-10 + >50% Plt fall -> **HIT**
- Warfarin skin necrosis -> think early **protein C depletion**

Last 5 minutes: Gplb-VWF adhesion; GplIb/IIIa-fibrinogen aggregation. PT=VII/common. aPTT=XII/XI/IX/VIII/common. Mix corrects deficiency. Hemophilia B=IX. DIC coag tests high; TTP normal. FVL resists protein C. HIT = stop heparin + non-heparin anticoagulant.

Primary source: Savage 50-slide coagulation deck. Official Week 7 objectives set emphasis. HIT is expanded from the uploaded companion. Modern Hemophilia B therapies are kept in the full notes as labeled external enrichment.